

42.510 Statistical Learning for Data Science

Introduction to Statistical Learning

Bikramjit Das

Week 1, Lecture 1

Section 1

Logistics

Lectures and tutorials

Lectures

- 2 x 2 hours weekly, partly theory and some R activity.
- Start on time, with 10-15 minute break in the middle.
- Please ask questions to clarify.

Tutorial

- 1 hour session
- Preliminaries
- R basics
- Exercises: practical and R-based
- Optional

Discussion

We will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the TA, and the instructor. Rather than emailing questions to the teaching staff, you are encouraged to post your questions on Piazza. <https://piazza.com/sutd.edu.sg/fall2025/42510>

Assessment

Assessment	Percentage	Period
Assignments	20%	throughout the term
Class participation	10%	throughout the term
Mid-term exam	25%	21st October
Final exam	25%	16th December
Project	20%	throughout the term

- Assignments are mainly for your benefit. Grades will be based on a few questions.
- Exams will be a mix of theory and practical (in R).
- Participation score will be based on attendance, responsiveness in class, occasional quiz.

Homework assignments

- Four homework assignments will be provided, roughly once every two to three weeks.
- Late submissions are allowed for partial credit. 25% of the total obtained points will be deducted for submitting within 48 hours of the deadline. Submissions after 48 hours will not be graded. Submit your assignments in pdf format via Gradescope.
- Programming-based assignments (using R) will not be graded; these are for your practice only. Solutions for these will be made available.
- Assignments can be accessed and submitted via e-dimension or through **Gradescope** using Entry Code: PGZE7D

Project

- ~ 3 members per team.
 - ▶ MTD-DS and non-MTD-DS separate teams (assigned by us).
- You can do one of the following:
 - ① Find an interesting data set and raise and answer meaningful questions.
 - ② Read an article on a Statistical learning topic, reproduce the results.
 - ③ Propose a Statistical learning method, exhibit on a dataset, discuss ...
- Milestones
 - ▶ Teams decided by 26th September
 - ▶ Preliminary report by 15th October (4 points)
 - ★ Background, methodology to be used, preliminary analysis (if any).
 - ★ 1 A4 page maximum, fontsize 12.
 - ▶ Presentation on 8th & 9th December (8 points)
 - ▶ Final report by 10th December (8 points)
 - ★ Like a scientific report/ journal paper. For example, it should include: abstract, introduction, methodology and analysis, conclusion.
 - ★ 6 A4 page maximum, fontsize 12.

Academic Integrity

Students are held accountable for the integrity of the work they submit. In particular, this means that for individual work (which includes homework assignment), copying and allowing others to copy are both prohibited. Working with others is fine and even encouraged – for instance, explaining to each other the meaning of a question or discussing how to approach a solution; however, you must write the solutions by yourself. Direct copying is an academic offence, and will result in a score of 0 for the assessment item.

Pre-requisites

Linear Algebra

- Vectors and matrices, multiplication
- Inner product, projection
- Orthogonality, linear dependence/ independence
- Eigenvalues and eigenvectors

Calculus

- Differential and integral calculus
- Multivariable and vector calculus
- Maxima, minima, optima

Some material is covered in MTD preliminary classes

Pre-requisites

Probability

- Probability distributions: joint and marginal, PDF, CDF
- Independence, conditional independence
- Random variables and expectations
- Covariance, covariance, and correlation matrices
- Multivariate Gaussian distribution
- Law of large numbers, central limit theorem

Some material is covered in MTD preliminary classes

Statistical inference (brief review: week 1, lecture 2)

- Basics of parametric and non-parametric estimation
- Maximum likelihood estimation
- Confidence intervals and hypothesis testing

Section 2

Introduction

Statistical Learning vs Machine learning

- Science of learning from data.
- Same intent but different history, emphasis, convention, etc.
- Related areas with high overlaps: analytics, data mining, artificial intelligence, deep learning, . . .
- *Statistical Learning* emphasizes uncertainty quantification, interpretation of models and estimates.
 - ▶ Questions are well-posed in general.
 - ▶ Lot of emphasis on survey, sampling, data collection, designing experiments.
- *Machine Learning* deals with many approaches for predictive modeling often employing black box or algorithmic techniques.
 - ▶ (almost) a consequence of information revolution: improvement in data processing and data storage.
 - ▶ huge amounts of data being available (of possibly low quality) but potentially valuable with information.

Are Machine Learning and Statistics Complementary? - Max Welling

Applications

- image recognition
- hand-written character recognition
- spam filtering
- recommendation systems
- autonomous vehicles
- market analysis
- healthcare analytics
- sports analytics
- smart manufacturing
- anomaly detection
- natural language processing
- generative AI
- ...

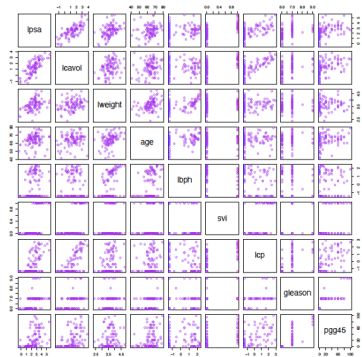
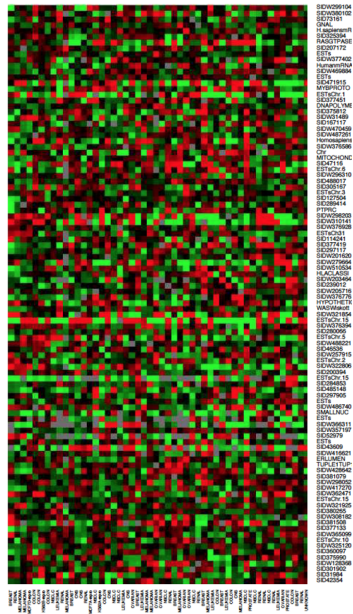
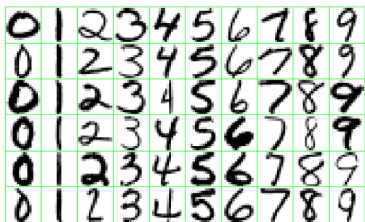


FIGURE 1.1. Scatterplot matrix of the prostate cancer data. The first row shows the response against each of the predictors in turn. Two of the predictors, svi and gleason, are categorical.

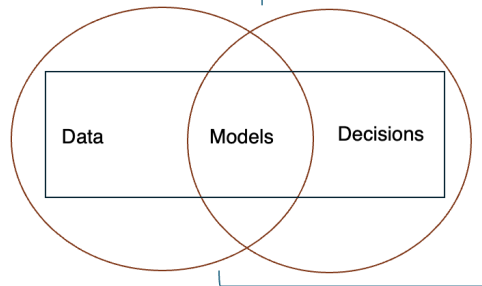


Data Science and Analytics ... an MTD perspective

Descriptive/ Predictive analytics:

Statistical learning + Machine learning

1. Data, Technology and Design (Term 1)
2. Statistical Learning for DS (Term 1)
3. Machine Learning and Analytics (Term 2)



Prescriptive analytics:

Optimization + Simulation

4. Optimization for DS (Term 2)
5. Digital twins in data-driven decisions (Term 3)

Data Science design project

5 Data science/analytics modules
+
2 Design modules

Overview of Statistical learning

Supervised learning

- Given measurement of certain *input variables* (or, *predictors*, or, *independent variables*, or, *features*) we want to predict *output variables* (or, *responses*, or, *dependent variables*).
- For *quantitative* output variables, we call the task **regression**.
 - ▶ e.g., predicting PSA (an antigen) using different prostate cancer measurements, predicting average house value in a city using various socio-economic factors of its neighborhood, etc.
- For *qualitative* output variables, we call the task **classification**.
 - ▶ e.g., handwritten digit recognition, email spam detection, etc.
- Comparison of methods via various performance measures (using Statistical Learning theory).

Overview of Statistical learning

Unsupervised learning

- Only *input variables* available (no output/ response).
- Find structure pattern in the data in order to cluster them, or find some latent characterization.
- No apparent measures of performance.
- Useful reducing dimension in high-dimensional problems (gene expressions), clustering of a large population (consumers for an online streaming service), etc.

Overview of Statistical learning

Other learning paradigms

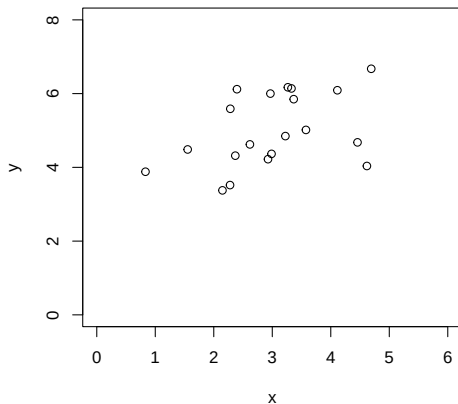
- **Semi-supervised learning:** few labeled and many unlabeled data points.
- **Active learning:** similar to semi-supervised, but users can choose which unlabeled points can be sent for labeling.
- **Online learning:** supervised method but not static. Data acquirement and predictions keep on happening on the fly.
- **Reinforcement learning:** here predictions may influence future input variables.
- **Generative modeling:** generate texts, images from “prompts” in an appropriate manner.

Overview of Statistical learning

Fairness, privacy and ethics

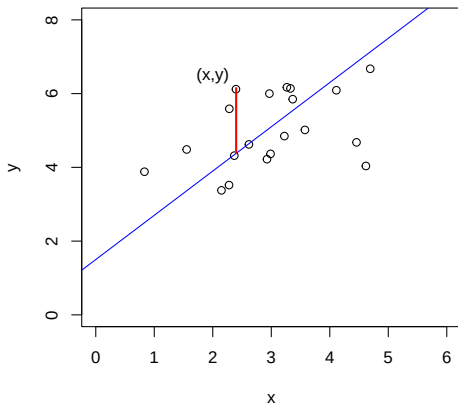
- The world of Data and AI has endless promises, one needs to be careful about potential dangers.
- Model/ training is just as good as the available data.
- Benefits of using data is often distributed unequally.
- Companies collect data on individuals.
 - ▶ is privacy protected? how much protection is there against leaks?
 - ▶ is consent properly taken? Is there a “right to forget”?
- Generative models use language, text, music created by authors, artists.
 - ▶ what is fair compensation?
 - ▶ is it possible to compensate at all and how?
- The risks of this unprecedented amount of data usage to make decisions is still quite poorly understood.

Learning: regression



- We have $n = 20$ points.
- $(x_1, y_1), \dots, (x_n, y_n)$.
- Find the best fitting line.

Learning: regression



- $(x_1, y_1), \dots, (x_{20}, y_{20})$.
- $D_i = |y_i - \beta_0 - \beta_1 x_i|$ where $(x_i, y_i) = (x, y)$

Regression: method of least squares

- Empirical Risk Minimization (ERM):

$$R_n(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n D_i^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- Find β_0, β_1 that minimizes $R_n(\beta_0, \beta_1)$: **method of least squares**.
- Let $\bar{x} = \sum_{i=1}^n x_i$ and $\bar{y} = \sum_{i=1}^n y_i$
- Solution (taking derivatives):

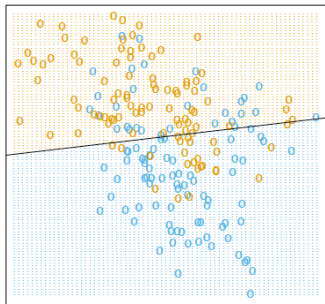
$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Regression: alternatives

- Least absolute deviation: minimize $\sum_{i=1}^n |D_i|$ w.r.t β_0, β_1 .
 - ▶ not analytically solvable.
 - ▶ many ways to solve including weighted least squares, polynomial time search, etc.
- Why linear?
 - ▶ polynomial, logarithmic, any function.
 - ▶ Fit $y = f(x) + \epsilon$.
 - ▶ decision trees, random forest, neural networks, etc.
- Why least squares (or least deviation)?
 - ▶ other metrics to quantify error: **loss function**.
 - ▶ e.g., relative error, root-mean squared error, etc.
 - ▶ e.g., robust case: Huber loss, quantile loss.
- Measuring and comparing performance.
 - ▶ out-of-sample error.
 - ▶ generalization error.
 - ▶ quantifying with statistical assumptions.

Learning problem: classification

Linear classifier



- Two classes. Class variable G : BLUE and ORANGE (simulated)
- Linear regression fit: $y_i = 0$ if color is BLUE, $y_i = 1$ if ORANGE.
- Fitted values are $\hat{Y}$. Predict class G with

$$\hat{G} = \begin{cases} \text{BLUE} & \text{if } \hat{Y} \leq 0.5, \\ \text{ORANGE} & \text{if } \hat{Y} > 0.5. \end{cases}$$

Learning problem: classification

Scenario 1

The (training) data in each class is generated from different bivariate Gaussian (normal) distributions which are uncorrelated and with different means.

- Linear decision boundary is perhaps the best we can do.
- Nearly optimal.
- Misclassification error will persist for future (test) data.

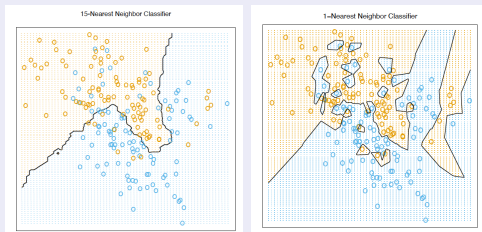
Scenario 2

The (training) data in each class is generated from a mixture of 10 Gaussian distributions. The means of the Gaussians are also Gaussian.

- Other non-linear models may work well.
- We need to be more careful about over-fitting.

Learning problem: classification

Nearest neighbor classifier



- For any point $x \in \mathbb{R}^2$ let $N_k(x)$ = set of k nearest neighbors of x .

$$\hat{Y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i.$$

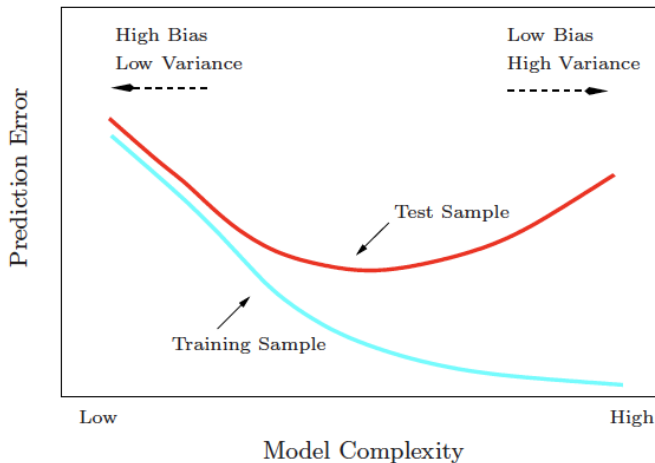
- Then we predict:

$$\hat{G}(x) = \begin{cases} \text{BLUE} & \text{if } \hat{Y}(x) \leq 0.5, \\ \text{ORANGE} & \text{if } \hat{Y}(x) > 0.5. \end{cases}$$

Classification: alternatives

- Generative vs Discriminative models
 - ▶ Linear discriminant analysis/ Quadratic discriminant analysis
 - ▶ Naive Bayes classifier
 - ▶ Multiclass classification
 - ▶ Logistic regression
 - ▶ Probit regression and link functions
- Other methods
 - ▶ Decision trees
 - ▶ Support vector machines
 - ▶ Nearest neighbor classifier
- Loss function
 - ▶ 0-1 Loss
 - ▶ Hinge loss
 - ▶ Cross-entropy loss
 - ▶ Multiclass: Softmax, KL loss

Model selection and overfitting



Rest of the week

- Darryl will do a refresher in R using R Studio for the first two weeks during the tutorial.
- Tuesday's class: *Preliminaries: Statistical inference*.
- Install R and R studio.
- Attempt doing the R Practice exercise (Assignment 0, this is not graded). Check that you can access this via Gradescope.

References

- Hastie et al. (2001), *The Elements of Statistical Learning*; Chapters 1,2.
- Bach (2024), *Learning Theory from First Principles*; Chapter 2.